Policy-Level Permutation Consistency in Small MARL
A Controlled Study of the Invariance-Coordination Mismatch

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Abstract

Teammate features in decentralized multi-agent reinforcement learning are often concatenated in an arbitrary fixed order, even when teammates are exchangeable. We study whether reducing this order sensitivity improves zero-shot coordination with unseen partners. Our testbed is *PICR-MA*, a lightweight policy-side regularizer that samples two independent permutations of teammate feature blocks and penalizes the symmetric Kullback–Leibler divergence between the resulting action distributions. PICR-MA leaves the inference-time policy architecture unchanged and adds only extra training-time forward passes. We evaluate PICR-MA against IPPO, Deep Sets, and Other-Play on a 5-seed, 50k-step PettingZoo MPE cohort (`simple_spread_v3`, `simple_adversary_v3`) using same-family held-out partner cross-play, permutation sensitivity, and runtime diagnostics; matrix-game sanity checks are retained only as appendix support. We also add a second matched 5-seed IPPO-versus-PICR-MA replication cohort at the same MPE budget. Across both cohorts, PICR-MA consistently lowers permutation sensitivity. The coordination effect is much less stable: the positive `simple_adversary_v3` cross-play delta remains wide in both cohorts, and the original negative `simple_spread_v3` delta reverses sign in the replication cohort. The main contribution of the paper is therefore not a new default MARL recipe, but a controlled empirical result: permutation-consistency diagnostics are easier to reproduce than held-out coordination gains.

1 Introduction

Zero-shot coordination (ZSC) asks whether an agent trained in self-play can cooperate with an unfamiliar partner at test time [Hu et al., 2020]. In decentralized MARL, one source of ZSC brittleness is mundane but pervasive: teammate features are commonly fed to a multilayer perceptron in a fixed concatenation order, so the policy may learn distinct roles for the "first" and "second" teammate even when the environment is symmetric. This creates an avoidable dependence on an arbitrary observation convention.

This paper studies a minimal intervention at the policy level. Rather than changing the policy architecture to a set encoder or graph network [Zaheer et al., 2017, Liu et al., 2020], we regularize the policy to produce similar action distributions under different teammate-order views of the same observation. The resulting method, PICR-MA, is intentionally simple: during training we apply two independent permutations to teammate feature blocks and penalize their action-distribution disagreement with a symmetric KL loss.

Preprint.

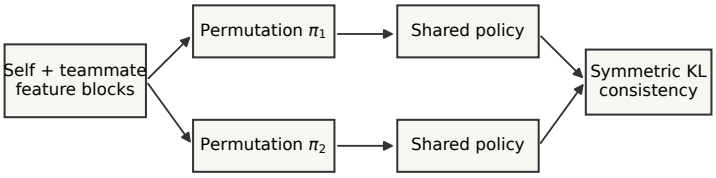


Figure 1: PICR-MA overview. For each training example, the teammate-specific observation blocks are permuted twice, the shared policy is evaluated on both views, and the resulting action distributions are matched with a symmetric KL penalty. Inference uses the original policy with no extra architectural machinery.

The interesting question is not only whether this penalty reduces a permutation-sensitivity diagnostic. The more important question is whether that diagnostic reduction improves downstream coordination with held-out partners. The answer in the current artifact is less stable than the diagnostic change itself. Across two matched 5-seed IPPO-versus-PICR-MA cohorts, the permutation-sensitivity shift is reproducible, while the coordination effect remains wide on `simple_adversary_v3` and even flips sign across cohorts on `simple_spread_v3`. This mismatch is the main scientific point of the workshop paper.

Contributions.

- We implement PICR-MA as a policy-side permutation-consistency regularizer for shared-parameter decentralized MARL with no inference-time architectural change.
- We build a controlled evaluation stack with held-out partner cross-play, explicit teammate-block slicing, permutation-sensitivity diagnostics, runtime tracking, and a second matched IPPO-versus-PICR-MA replication cohort for the main MPE tasks.
- We show that PICR-MA reliably reduces permutation sensitivity across both matched MPE cohorts, but its effect on held-out coordination is materially less stable: positive but wide on `simple_adversary_v3`, and sign-flipping across cohorts on `simple_spread_v3`.
- We identify a concrete diagnostic failure mode for this family of ideas: lower teammate-order sensitivity does not by itself imply better zero-shot coordination.

2 Method

Each agent observation is decomposed into non-teammate features s and teammate-specific blocks $U = [u_1, \ldots, u_{n-1}]$. For two independently sampled teammate permutations π_1 and π_2, we form

$$o_{(1)} = [s \| \pi_1(U)], \qquad p_1 = \pi_\theta(\cdot \mid o_{(1)}), \tag{1}$$

$$o_{(2)} = [s \| \pi_2(U)], \qquad p_2 = \pi_\theta(\cdot \mid o_{(2)}). \tag{2}$$

The regularizer is

$$\mathcal{L}_{\text{PICR-MA}} = \lambda \, \mathbb{E}\Big[\tfrac{1}{2}\mathrm{KL}(p_1 \| p_2) + \tfrac{1}{2}\mathrm{KL}(p_2 \| p_1)\Big]. \tag{3}$$

The training objective is $\mathcal{L}_{\text{RL}} + \mathcal{L}_{\text{PICR-MA}}$, where $\mathcal{L}_{\text{RL}}$ is the base on-policy objective. The implementation keeps the base MLP policy unchanged at inference time; the only overhead is the additional training-time forward passes required for the consistency term.

We also implement teammate dropout as an ablation that zeros a random subset of teammate blocks during training. It is not part of the core method and, empirically, it does not strengthen the story.

3 Experimental Setup

Environments. The main evidence comes from two PettingZoo MPE tasks [Terry et al., 2021]: `simple_spread_v3`, a fully cooperative three-agent coverage task, and `simple_adversary_v3`, where a cooperative team interacts with an adversary. Matrix coordination and stag-hunt variants are retained as fast sanity checks and are reported in the appendix rather than driving the main claim.

Baselines. We compare four methods under matched training budgets: IPPO, IPPO+PICR-MA, IPPO with a Deep Sets encoder, and Other-Play-style teammate-order randomization [Schulman et al., 2017, Terry et al., 2020, Zaheer et al., 2017, Hu et al., 2020]. The comparison is intentionally practical: Deep Sets provides an architectural invariance baseline, and Other-Play provides a symmetry-randomization baseline with a stronger ZSC pedigree.

Main cohort and pilot ablations. The main MPE cohort uses 5 seeds, 50k environment steps per seed, and held-out partner evaluation from independently trained policies sharing the same run tag. A 20k, 3-seed pilot sweep selected $\lambda \in \{0.02, 0.05, 0.10\}$ for PICR-MA and evaluated teammate dropout at probability 0.1. The main 50k PICR-MA setting uses $\lambda = 0.02$ because it gave the strongest held-out cross-play among the pilot settings, even though larger λ values reduced JS more aggressively.

Evaluation. For each run we record training return, held-out partner cross-play return, permutation sensitivity measured by Jensen–Shannon divergence under teammate-order swaps, and environment-specific diagnostics such as collisions or goal-margin statistics. The headline cross-play number in the main table is same-family held-out cross-play, meaning the ego policy is evaluated against independently trained partners from the same method family and run tag; cross-family partners are reported separately in the appendix. Cross-play results are produced from saved checkpoints with a partner-pool filter that excludes the evaluated run itself and respects the intended run tag. To test whether the original matched-seed pattern was stable, we also train a second 5-seed IPPO-versus-PICR-MA cohort at the same 50k MPE budget and analyze it separately rather than merging it into the baseline-comparison table.

Implementation and compute. All experiments were executed sequentially on a single Apple Silicon laptop under the repo's lean-safe resource profile. The current learner is a small NumPy actor–critic rather than a full PyTorch PPO stack; this keeps the artifact compact and reproducible, but it also limits external ambition and informs the workshop-scale framing of the paper. IPPO, PICR-MA, and Other-Play share the same MLP backbone and optimization budget, so PICR-MA's extra cost comes from two additional training-time forward passes rather than a larger inference model; only Deep Sets changes capacity (1606 versus 3302 cooperative parameters on `simple_spread_v3`, and 1350 versus 3238 on `simple_adversary_v3`).

4 Results

Permutation consistency is real. Table 1 and Figure 4 show that PICR-MA does what it is supposed to do mechanically. In the main 50k cohort, it lowers teammate-order sensitivity relative to IPPO on both MPE tasks, from 0.0263 ± 0.0063 to 0.0127 ± 0.0019 on `simple_spread_v3` and from 0.0102 ± 0.0066 to 0.0053 ± 0.0025 on `simple_adversary_v3`. Deep Sets drives this diagnostic to zero, as expected for an architectural invariance baseline.

Lower sensitivity does not reliably imply better coordination. Figure 3 overlays the original matched IPPO→PICR-MA seed pairs with the second 5-seed replication cohort, while Table 5 gives the corresponding bootstrap summaries. In the original cohort, PICR-MA lowers JS in all five `simple_spread_v3` pairs and loses on same-family held-out cross-play in all five; the mean cross-play delta is -1.310 with a 95% bootstrap interval of $[-2.145, -0.779]$ after the higher-episode reevaluation. In the replication cohort, PICR-MA again lowers JS on `simple_spread_v3`, but now improves same-family held-out cross-play in all five pairs, with mean delta $+3.011$ and interval $[2.603, 3.467]$. Pooling the two cohorts gives a 10-pair `simple_spread_v3` cross-play delta of $+0.851$ with interval $[-0.582, 2.233]$, while the pooled ΔJS remains clearly negative at -0.014 with interval $[-0.023, -0.006]$. On `simple_adversary_v3`, both cohorts remain positive on average

Table 1: Main 50k-step, 5-seed MPE results. The cross-play column reports same-family held-out partner cross-play; lower JS is better. The table makes the workshop framing explicit: in the original baseline-comparison cohort, PICR-MA lowers order sensitivity, improves IPPO on `simple_adversary_v3`, and loses on `simple_spread_v3`.

Environment	Method	Train return	Same-family cross-play	JS ↓	Env metric
Simple Adversary	IPPO	2.346 ± 1.813	16.494 ± 3.602	0.010 ± 0.007	margin=0.260
Simple Adversary	PICR-MA	6.585 ± 3.502	19.657 ± 2.352	0.005 ± 0.003	margin=0.345
Simple Adversary	Deep Sets	7.965 ± 1.698	11.357 ± 3.121	0.000 ± 0.000	margin=0.392
Simple Adversary	Other-Play	3.603 ± 1.497	20.605 ± 3.195	0.014 ± 0.007	margin=0.310
Simple Spread	IPPO	-21.952 ± 0.472	-25.644 ± 0.268	0.026 ± 0.006	coll=0.164
Simple Spread	PICR-MA	-23.425 ± 0.625	-26.954 ± 0.536	0.013 ± 0.002	coll=0.155
Simple Spread	Deep Sets	-24.776 ± 1.413	-25.436 ± 0.338	0.000 ± 0.000	coll=0.065
Simple Spread	Other-Play	-21.626 ± 0.209	-25.583 ± 0.280	0.025 ± 0.013	coll=0.079

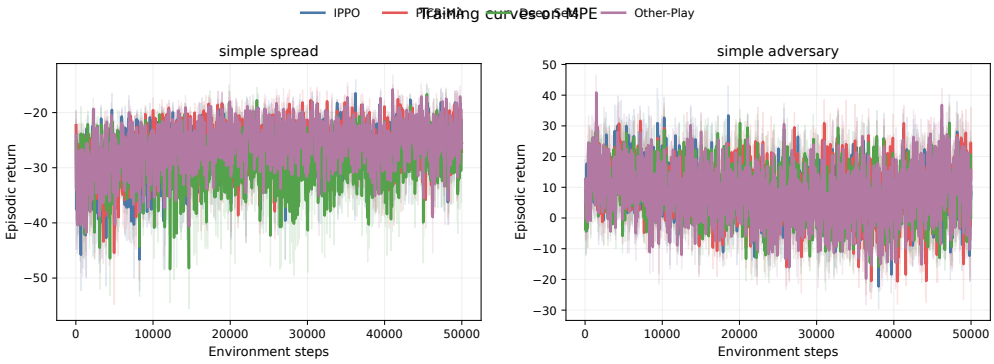


Figure 2: MPE learning curves for the main 50k cohort. The curves show a genuine training signal rather than a toy smoke test, but they also make the scale limitation visible: `simple_spread_v3` is still improving late in training.

for same-family held-out cross-play ($+3.163$ and $+2.447$), but both intervals remain wide and the pooled 10-pair cross-play interval still spans zero ($[-2.929, 8.842]$). The most reproducible signal is therefore the diagnostic shift, not the coordination gain.

Baselines matter. The baseline story is also mixed. Deep Sets is the strongest invariance baseline but is weak on `simple_adversary_v3` held-out cross-play, suggesting that exact invariance alone can remove useful asymmetries in that environment. Other-Play is slightly stronger than PICR-MA on `simple_adversary_v3` and stronger than the original PICR-MA cohort on `simple_spread_v3`. The appendix cross-family breakdown also shows that PICR-MA's positive adversary signal is concentrated in same-family partners rather than a uniform gain across partner families. The honest takeaway is therefore not that PICR-MA is a better default than architectural invariance or role randomization, but that it occupies a small simplicity–performance niche whose coordination effect is too unstable to summarize with a single environment-level sign.

Ablations and overhead. The 20k pilot ablations in Table 2 show that stronger regularization does not monotonically help. On `simple_spread_v3`, increasing λ reduces JS but steadily worsens held-out cross-play. On `simple_adversary_v3`, the mild setting $\lambda = 0.02$ gives the best held-out cross-play in the sweep. Teammate dropout harms both tasks. Runtime overhead is moderate rather than negligible: in the main 50k cohort, PICR-MA costs $1.62\times$ IPPO on `simple_spread_v3` and $1.33\times$ IPPO on `simple_adversary_v3`, while remaining well below the Deep Sets cost on `simple_spread_v3`.

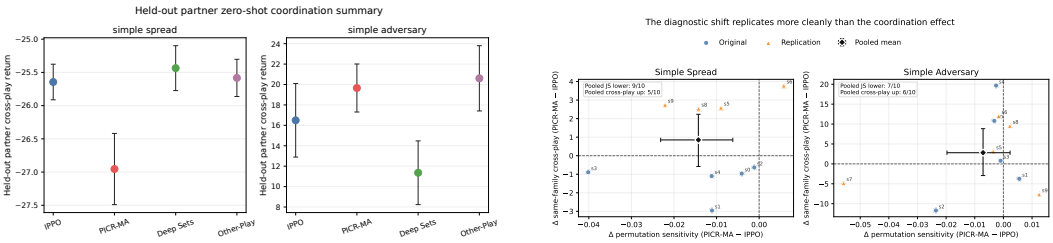


Figure 3: Left: same-family held-out partner ZSC summary for the original baseline-comparison cohort. Right: matched IPPO→PICR-MA seed-pair deltas across the original and replication 5-seed cohorts. In both environments the permutation-sensitivity shift trends leftward more consistently than the coordination shift trends upward. The black diamond shows the pooled 10-pair bootstrap mean and interval.

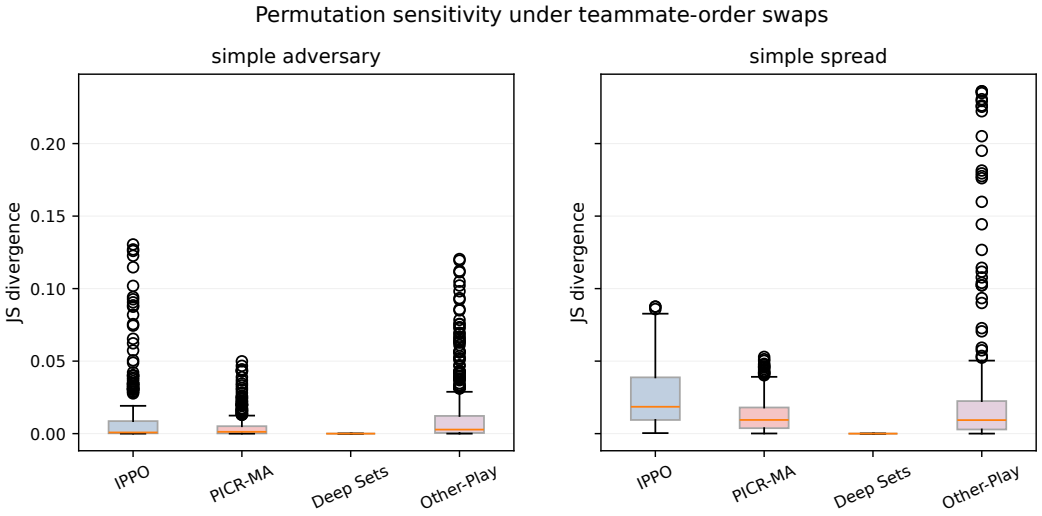


Figure 4: Permutation sensitivity distributions over saved evaluation states. Deep Sets is exactly invariant by construction; PICR-MA improves substantially over IPPO but not always in the way that matters most for held-out coordination.

5 Discussion

This paper should not be read as evidence that policy-level permutation consistency is a new default recipe for ZSC in MARL. The stronger and more interesting conclusion is narrower: it is possible to reduce order sensitivity substantially with a very small policy-side intervention, but the downstream value of that intervention is less stable than the diagnostic change it induces.

The contrast between the two MPE tasks is now suggestive rather than decisive. In the original cohort, `simple_spread_v3` looked clearly negative and `simple_adversary_v3` modestly positive; the second matched cohort preserves the adversary-average positivity but flips the spread sign while keeping the diagnostic shift intact. That makes the deeper lesson sharper. Invariance diagnostics measure one property of the policy, and that property can replicate even when the downstream coordination effect remains fragile to the particular seed cohort and partner pool. Lower teammate-order sensitivity is therefore informative, but it is not a reliable proxy for zero-shot coordination quality.

Limitations. The learner is a compact NumPy implementation rather than a standard PyTorch PPO codebase. The main MPE cohort is stronger than the original smoke runs, but it remains a 50k-step laptop-scale study rather than a large benchmark campaign. Matrix games are useful sanity checks, not headline evidence. These limitations are acceptable for a focused workshop submission, but they should prevent stronger claims than the ones made here.

Table 2: PICR-MA pilot ablations from the 20k, 3-seed sweep. The cross-play column reports same-family held-out partner cross-play. The table supports a narrow claim: mild regularization is preferable, and teammate dropout is not helpful in this artifact.

Environment	Setting	Same-family cross-play	JS ↓	Runtime (s)
Simple Spread	$\lambda = 0.02$	-31.673 ± 0.520	0.025 ± 0.006	35.936
Simple Spread	$\lambda = 0.05$	-33.884 ± 1.281	0.022 ± 0.004	55.688
Simple Spread	$\lambda = 0.10$	-34.673 ± 1.060	0.009 ± 0.002	91.293
Simple Spread	dropout $p = 0.10$	-37.486 ± 2.504	0.016 ± 0.004	14.812
Simple Adversary	$\lambda = 0.02$	25.743 ± 1.344	0.016 ± 0.006	50.579
Simple Adversary	$\lambda = 0.05$	22.688 ± 3.005	0.012 ± 0.002	66.191
Simple Adversary	$\lambda = 0.10$	24.743 ± 2.973	0.011 ± 0.003	39.520
Simple Adversary	dropout $p = 0.10$	16.897 ± 2.086	0.026 ± 0.016	18.596

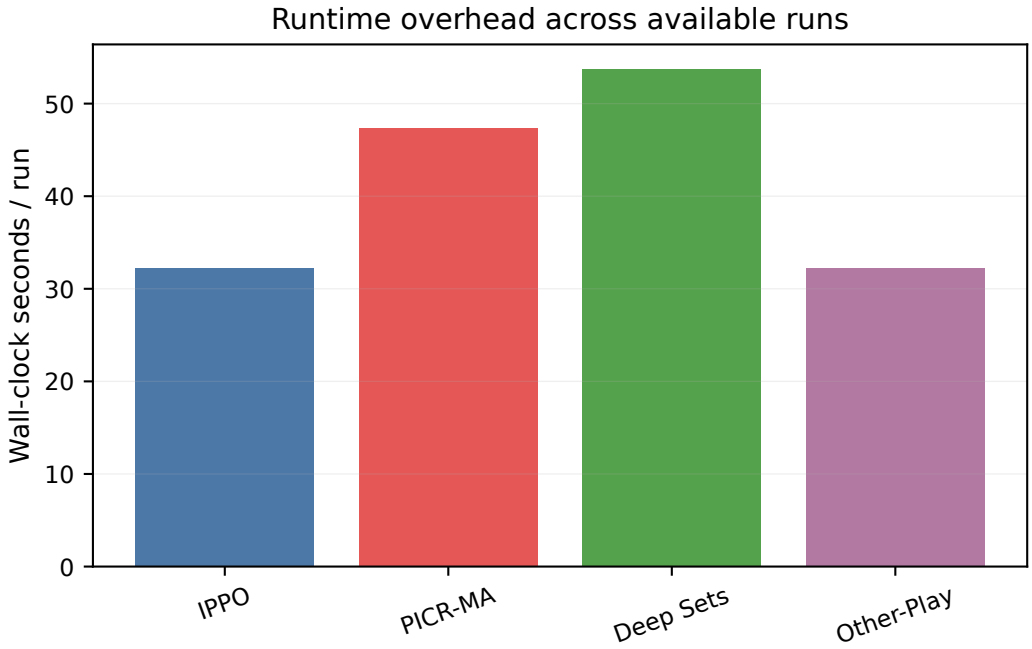


Figure 5: Per-run wall-clock runtime across the main cohort. PICR-MA is meaningfully more expensive than IPPO, but still simpler than an architectural rewrite and cheaper than Deep Sets on `simple_spread_v3`.

6 Conclusion

PICR-MA is a clean way to inject permutation consistency into decentralized MARL without changing the inference-time policy. In the current controlled study, that bias reliably lowers teammate-order sensitivity across two matched MPE cohorts, but the paired-seed coordination effect remains unstable: it stays wide on `simple_adversary_v3` and even flips sign across `simple_spread_v3` cohorts. For workshop purposes, the more durable contribution is the empirical lesson: lower permutation sensitivity is reproducible here, but it is insufficient as a standalone proxy for zero-shot coordination quality.

Acknowledgments and Disclosure of Funding

This project used only local laptop-scale compute and open-source dependencies. No external experiment tracker, paid API, or cloud compute service was used in the reported results.

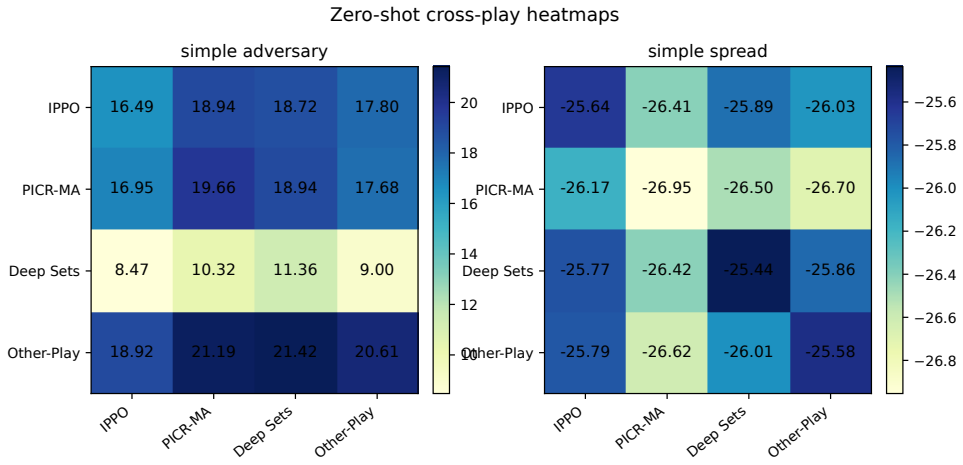


Figure 6: Held-out partner cross-play heatmaps from the main 50k cohort. The main paper uses a compact summary plot; this appendix figure exposes the full family-by-family structure.

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A Additional Cross-Play Results

Figure 6 shows the full cross-play matrices behind the main-paper summary plot. The matrices make two points visible. First, `simple_spread_v3` remains broadly difficult for PICR-MA, not just on the diagonal. Second, on `simple_adversary_v3`, the PICR-MA advantage over IPPO is concentrated in same-family partners rather than a uniform cross-family gain, while Other-Play remains slightly stronger overall.

B Additional Tables

C Matrix-Game Sanity Checks

The matrix games remain useful as fast debugging environments for observation slicing, partner-pool evaluation, and rollout correctness, but they do not determine the paper's main claim. We therefore treat them as supporting evidence rather than headline evidence.

Table 3: Held-out partner cross-play split into same-family and cross-family partners.

Environment	Ego method	Same-family	Cross-family
Simple Adversary	Deep Sets	11.357 $\pm$ 2.055	9.264 $\pm$ 1.086
Simple Adversary	IPPO	16.494 $\pm$ 2.485	18.485 $\pm$ 1.368
Simple Adversary	Other-Play	20.605 $\pm$ 2.552	20.510 $\pm$ 1.120
Simple Adversary	PICR-MA	19.657 $\pm$ 1.951	17.855 $\pm$ 0.878
Simple Spread	Deep Sets	-25.436 $\pm$ 0.639	-26.016 $\pm$ 0.263
Simple Spread	IPPO	-25.644 $\pm$ 0.302	-26.109 $\pm$ 0.259
Simple Spread	Other-Play	-25.583 $\pm$ 0.416	-26.137 $\pm$ 0.293
Simple Spread	PICR-MA	-26.954 $\pm$ 0.446	-26.459 $\pm$ 0.298

Table 4: Matched-seed PICR-MA versus IPPO deltas. The cross-play rows report same-family held-out partner cross-play, and negative ΔJS is desirable. The table makes the workshop claim explicit: the diagnostic shift is more consistent than the coordination shift.

Environment	Quantity	Mean delta	95% bootstrap CI	Direction count
Simple Spread	Same-family cross-play	-1.310	[-2.145, -0.779]	0/5 higher
Simple Spread	JS $\downarrow$	-0.014	[-0.027, -0.004]	5/5 lower
Simple Adversary	Same-family cross-play	3.163	[-6.021, 13.211]	3/5 higher
Simple Adversary	JS $\downarrow$	-0.005	[-0.015, 0.002]	4/5 lower

D Observation Slicing, Hyperparameters, and Assets

Observation slicing. Teammate-block indices are explicitly cached per environment rather than inferred implicitly at each update. For `simple_adversary_v3`, the consistency loss is applied only to the cooperative team. The precise slicing assumptions and any environment-specific simplifications are recorded in `results/OVERRIDES.md`.

Hyperparameters. The main reported learner uses hidden size 64, learning rate 3×10^{-3}, discount 0.99, GAE parameter 0.95, PPO clip ratio 0.2, and four update epochs per rollout. The main PICR-MA cohort uses $\lambda = 0.02$ and no teammate dropout. Exact commands, profiles, and artifact inventories are listed in `results/FINAL_REPORT.md`.

Model capacity and fairness. IPPO, PICR-MA, and Other-Play share the same inference-time MLP and optimizer settings; only Deep Sets changes the encoder capacity. In the main cohort this corresponds to 1606 cooperative parameters for IPPO/PICR-MA/Other-Play versus 3302 for Deep Sets on `simple_spread_v3`, and 1350 versus 3238 on `simple_adversary_v3`. PICR-MA's extra runtime therefore comes from training-time consistency passes rather than a larger deployed policy.

Assets and licenses. The experiments rely on PettingZoo 1.24.3, Gymnasium 1.2.3, and SuperSuit 3.10.0. Local package metadata reports MIT licenses for these core environment assets. All reported figures and tables are generated from files stored under `results/`.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the paper's scope narrowly and match the empirical evidence: PICR-MA lowers permutation sensitivity, helps on one MPE task, hurts on another, and does not support a universal ZSC claim.

   Guidelines:

Table 5: Two matched IPPO-versus-PICR-MA cohorts at the same 50k MPE budget. The replication cohort preserves the diagnostic shift but does not preserve a stable coordination sign.

Environment	Cohort	Quantity	Mean delta	95% bootstrap CI
Simple Spread	Original 5-seed	Same-family cross-play	-1.310	[-2.145, -0.779]
Simple Spread	Original 5-seed	JS ↓	-0.014	[-0.027, -0.004]
Simple Spread	Replication 5-seed	Same-family cross-play	3.011	[2.603, 3.467]
Simple Spread	Replication 5-seed	JS ↓	-0.015	[-0.027, -0.003]
Simple Spread	Pooled 10-pair	Same-family cross-play	0.851	[-0.582, 2.233]
Simple Spread	Pooled 10-pair	JS ↓	-0.014	[-0.023, -0.006]
Simple Adversary	Original 5-seed	Same-family cross-play	3.163	[-6.021, 13.211]
Simple Adversary	Original 5-seed	JS ↓	-0.005	[-0.015, 0.002]
Simple Adversary	Replication 5-seed	Same-family cross-play	2.447	[-4.357, 9.251]
Simple Adversary	Replication 5-seed	JS ↓	-0.009	[-0.034, 0.006]
Simple Adversary	Pooled 10-pair	Same-family cross-play	2.805	[-2.929, 8.842]
Simple Adversary	Pooled 10-pair	JS ↓	-0.007	[-0.020, 0.002]

Table 6: Permutation sensitivity summary across environments and methods.

Environment	Method	JS divergence
Simple Adversary	Deep Sets	0.000 ± 0.000
Simple Adversary	IPPO	0.010 ± 0.001
Simple Adversary	Other-Play	0.014 ± 0.001
Simple Adversary	PICR-MA	0.005 ± 0.001
Simple Spread	Deep Sets	0.000 ± 0.000
Simple Spread	IPPO	0.026 ± 0.001
Simple Spread	Other-Play	0.025 ± 0.003
Simple Spread	PICR-MA	0.013 ± 0.001

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

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Table 7: Runtime summary relative to IPPO.

Environment	Method	Runtime (s)	Relative to IPPO
Simple Adversary	Deep Sets	44.885	1.34×
Simple Adversary	IPPO	33.504	1.00×
Simple Adversary	Other-Play	32.045	0.96×
Simple Adversary	PICR-MA	44.575	1.33×
Simple Spread	Deep Sets	62.529	2.02×
Simple Spread	IPPO	31.004	1.00×
Simple Spread	Other-Play	32.329	1.04×
Simple Spread	PICR-MA	50.235	1.62×

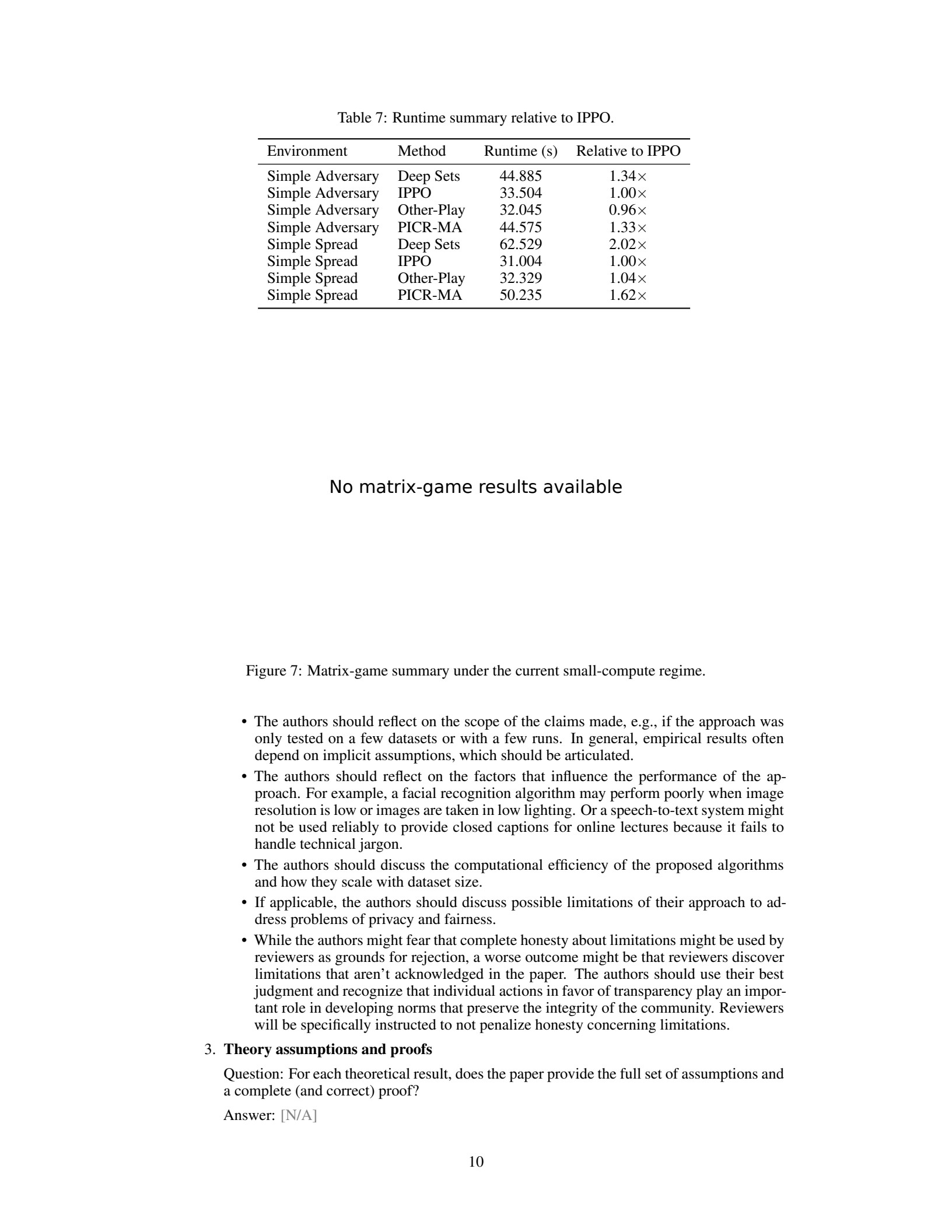


Figure 7: Matrix-game summary under the current small-compute regime.

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